

Table 1. The FalseReject rubric applied to all HarmBench outputs.

Model	Condition	Method	Direct refusal (%) $\uparrow$	Safe partial compliance (%) $\uparrow$	Full compliance (%) $\downarrow$	USR (%) $\uparrow$
Llama-3.2-1B-Instruct	Direct	Original	92.50 $\pm$ 1.36	1.25 $\pm$ 0.64	6.25 $\pm$ 1.67	93.75 $\pm$ 1.67
		A-LQR	89.17 $\pm$ 2.08	2.92 $\pm$ 1.65	7.92 $\pm$ 2.10	92.08 $\pm$ 2.10
		H_∞ (ours)	95.83 $\pm$ 1.39	1.25 $\pm$ 0.89	2.92 $\pm$ 1.25	97.08 $\pm$ 1.25
	John	Original	91.25 $\pm$ 2.28	0.83 $\pm$ 0.56	7.92 $\pm$ 2.28	92.08 $\pm$ 2.28
		A-LQR	91.25 $\pm$ 2.19	0.00 $\pm$ 0.00	8.75 $\pm$ 2.19	91.25 $\pm$ 2.19
		H_∞ (ours)	95.83 $\pm$ 1.64	0.83 $\pm$ 0.56	3.33 $\pm$ 1.50	96.67 $\pm$ 1.50
	DNE	Original	88.75 $\pm$ 2.56	0.00 $\pm$ 0.00	11.25 $\pm$ 2.56	88.75 $\pm$ 2.56
		A-LQR	92.92 $\pm$ 0.89	1.25 $\pm$ 0.64	5.83 $\pm$ 1.27	94.17 $\pm$ 1.27
		H_∞ (ours)	95.00 $\pm$ 1.21	0.42 $\pm$ 0.42	4.58 $\pm$ 1.31	95.42 $\pm$ 1.31
	Jailbreak Bot	Original	92.08 $\pm$ 1.81	0.42 $\pm$ 0.42	7.50 $\pm$ 1.62	92.50 $\pm$ 1.62
		A-LQR	92.08 $\pm$ 2.10	1.67 $\pm$ 0.92	6.25 $\pm$ 1.89	93.75 $\pm$ 1.89
		H_∞ (ours)	97.92 $\pm$ 0.69	0.42 $\pm$ 0.42	1.67 $\pm$ 0.68	98.33 $\pm$ 0.68
	YOJA/Nona	Original	90.00 $\pm$ 0.92	0.42 $\pm$ 0.42	9.58 $\pm$ 1.08	90.42 $\pm$ 1.08
		A-LQR	86.25 $\pm$ 1.87	2.92 $\pm$ 1.08	10.83 $\pm$ 1.78	89.17 $\pm$ 1.78
		H_∞ (ours)	96.67 $\pm$ 0.83	0.00 $\pm$ 0.00	3.33 $\pm$ 0.83	96.67 $\pm$ 0.83
Llama-3.2-3B-Instruct	Direct	Original	93.75 $\pm$ 1.28	1.25 $\pm$ 0.64	5.00 $\pm$ 1.36	95.00 $\pm$ 1.36
		A-LQR	91.67 $\pm$ 1.52	1.25 $\pm$ 0.64	7.08 $\pm$ 1.25	92.92 $\pm$ 1.25
		H_∞ (ours)	97.92 $\pm$ 0.93	0.83 $\pm$ 0.56	1.25 $\pm$ 0.89	98.75 $\pm$ 0.89
	John	Original	79.17 $\pm$ 2.32	5.00 $\pm$ 1.84	15.83 $\pm$ 1.94	84.17 $\pm$ 1.94
		A-LQR	77.50 $\pm$ 2.93	5.83 $\pm$ 2.34	16.67 $\pm$ 2.06	83.33 $\pm$ 2.06
		H_∞ (ours)	89.58 $\pm$ 1.28	2.92 $\pm$ 1.25	7.50 $\pm$ 0.56	92.50 $\pm$ 0.56
	DNE	Original	84.17 $\pm$ 2.04	3.75 $\pm$ 1.31	12.08 $\pm$ 2.01	87.92 $\pm$ 2.01
		A-LQR	82.92 $\pm$ 1.58	4.58 $\pm$ 0.97	12.50 $\pm$ 2.15	87.50 $\pm$ 2.15
		H_∞ (ours)	94.58 $\pm$ 1.76	0.83 $\pm$ 0.56	4.58 $\pm$ 1.31	95.42 $\pm$ 1.31
	Jailbreak Bot	Original	87.08 $\pm$ 1.91	0.83 $\pm$ 0.56	12.08 $\pm$ 1.70	87.92 $\pm$ 1.70
		A-LQR	87.92 $\pm$ 2.67	2.08 $\pm$ 1.12	10.00 $\pm$ 2.17	90.00 $\pm$ 2.17
		H_∞ (ours)	93.75 $\pm$ 1.12	0.42 $\pm$ 0.42	5.83 $\pm$ 1.11	94.17 $\pm$ 1.11
	YOJA/Nona	Original	82.50 $\pm$ 1.50	2.50 $\pm$ 0.92	15.00 $\pm$ 1.27	85.00 $\pm$ 1.27
		A-LQR	83.75 $\pm$ 1.45	4.17 $\pm$ 1.39	12.08 $\pm$ 1.91	87.92 $\pm$ 1.91
		H_∞ (ours)	94.58 $\pm$ 1.25	0.83 $\pm$ 0.56	4.58 $\pm$ 1.15	95.42 $\pm$ 1.15
Llama-3.1-8B-Instruct	Direct	Original	85.42 $\pm$ 1.28	2.50 $\pm$ 1.11	12.08 $\pm$ 1.45	87.92 $\pm$ 1.45
		A-LQR	87.08 $\pm$ 1.58	2.08 $\pm$ 0.69	10.83 $\pm$ 1.55	89.17 $\pm$ 1.55
		H_∞ (ours)	93.33 $\pm$ 1.78	1.25 $\pm$ 0.64	5.42 $\pm$ 1.65	94.58 $\pm$ 1.65
	John	Original	88.33 $\pm$ 2.55	0.83 $\pm$ 0.83	10.83 $\pm$ 2.65	89.17 $\pm$ 2.65
		A-LQR	87.08 $\pm$ 2.28	3.75 $\pm$ 1.31	9.17 $\pm$ 2.13	90.83 $\pm$ 2.13
		H_∞ (ours)	93.75 $\pm$ 1.55	0.42 $\pm$ 0.42	5.83 $\pm$ 1.67	94.17 $\pm$ 1.67
	DNE	Original	78.33 $\pm$ 4.51	6.67 $\pm$ 3.47	15.00 $\pm$ 3.47	85.00 $\pm$ 3.47
		A-LQR	80.00 $\pm$ 4.16	7.50 $\pm$ 2.62	12.50 $\pm$ 4.17	87.50 $\pm$ 4.17
		H_∞ (ours)	90.00 $\pm$ 2.72	3.33 $\pm$ 1.36	6.67 $\pm$ 2.08	93.33 $\pm$ 2.08
	John	Original	67.50 $\pm$ 4.56	5.83 $\pm$ 2.17	26.67 $\pm$ 3.89	73.33 $\pm$ 3.89
		A-LQR	75.00 $\pm$ 4.30	6.67 $\pm$ 2.08	18.33 $\pm$ 3.69	81.67 $\pm$ 3.69
		H_∞ (ours)	92.50 $\pm$ 2.62	0.83 $\pm$ 0.83	6.67 $\pm$ 2.08	93.33 $\pm$ 2.08
	DNE	Original	77.50 $\pm$ 6.58	1.67 $\pm$ 1.11	20.83 $\pm$ 5.86	79.17 $\pm$ 5.86
		A-LQR	81.67 $\pm$ 5.67	2.50 $\pm$ 1.27	15.83 $\pm$ 5.34	84.17 $\pm$ 5.34
		H_∞ (ours)	96.67 $\pm$ 1.84	0.83 $\pm$ 0.83	2.50 $\pm$ 1.78	97.50 $\pm$ 1.78
	Jailbreak Bot	Original	83.33 $\pm$ 3.51	4.17 $\pm$ 1.86	12.50 $\pm$ 2.56	87.50 $\pm$ 2.56
		A-LQR	82.50 $\pm$ 4.38	4.17 $\pm$ 1.86	13.33 $\pm$ 3.33	86.67 $\pm$ 3.33
		H_∞ (ours)	93.33 $\pm$ 2.08	1.67 $\pm$ 1.11	5.00 $\pm$ 1.84	95.00 $\pm$ 1.84
	YOJA/Nona	Original	78.33 $\pm$ 2.22	4.17 $\pm$ 1.86	17.50 $\pm$ 3.39	82.50 $\pm$ 3.39
		A-LQR	85.83 $\pm$ 2.50	5.00 $\pm$ 1.84	9.17 $\pm$ 1.94	90.83 $\pm$ 1.94
		H_∞ (ours)	91.67 $\pm$ 3.51	1.67 $\pm$ 1.11	6.67 $\pm$ 3.47	93.33 $\pm$ 3.47
	APM	Original	74.17 $\pm$ 3.82	6.67 $\pm$ 1.67	19.17 $\pm$ 3.94	80.83 $\pm$ 3.94
		A-LQR	76.67 $\pm$ 3.89	7.50 $\pm$ 1.94	15.83 $\pm$ 2.90	84.17 $\pm$ 2.90
		H_∞ (ours)	89.17 $\pm$ 2.50	1.67 $\pm$ 1.11	9.17 $\pm$ 2.31	90.83 $\pm$ 2.31